

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$u_i^{35,35}(y, z) = \left\{ \begin{array}{ll} -5b_i^{35,35}(y, z) & if \\ -1 & if \\ 1 & if \end{array} \quad \begin{array}{l} |5b_i^{35,35}(y, z)| \leq 1, \\ -5b_i^{35,35}(y, z) < -1, \\ 5b_i^{35,35}(y, z) > 1, \end{array} \right\}, \quad i = 1, 2,$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$(VII)' \quad \Psi_{\mathbf{D}} = \begin{pmatrix} a_1 + \mathbf{j}b_1 \\ a_2 + \mathbf{j}b_2 \\ a_3 + \mathbf{j}b_3 \\ a_4 + \mathbf{j}b_4 \end{pmatrix} \leftrightarrow \Psi_{\mathbf{C}}^{SR} = \begin{pmatrix} -b_1 + \mathbf{i}b_3 \\ -b_2 + \mathbf{i}b_4 \\ -a_1 + \mathbf{i}a_3 \\ -a_2 + \mathbf{i}a_4 \end{pmatrix}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{ll} A) & J_1, J_2, J_3 \text{ unrestricted,} \quad l \geq 6 + \frac{1}{2}(J_1 + 2J_2 + 3J_3) + 2(a_1 + a_2) \\ B) & J_3 = 0, \quad l = 4 + \frac{1}{2}(J_1 + 2J_2) + 2(a_1 + a_2) \\ C) & J_2 = J_3 = 0, \quad l = 2 + \frac{1}{2}J_1 + 2(a_1 + a_2) \\ D) & J_1 = J_2 = J_3 = 0, \quad l = 2(a_1 + a_2) \end{array}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & N_{L_n}^d \mathbb{E} \left(Tr(E_{H_{C_p}^\omega}(J_{L_n, E})) (Tr(E_{H_{C_p}^\omega}(J_{L_n, E}) - 1) \right) \\ & \leq N_{L_n}^d \left(Q_\mu(|J_{L_n, E}|) |C_p| \right)^2 \\ & \leq N_{L_n}^d (|J_{L_n, E}|^\alpha |C_p|) ^2 \\ & = O \left(\beta_{L_n}^{-2\alpha} N_{L_n}^d \left(\frac{2L_n + 1}{N_{L_n}} \right)^{2d} \right). \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\sum_j q_a^j(|X_j|^2 - |Y_j|^2)\eta_j^+\bar{\eta}_j^+ = r_a^2\eta_a^+\bar{\eta}_a^+ \quad (a)$$

$$\sum_j q_a^j(X_j\bar{Y}_j)\eta_j^+\eta_j^+ = 0 \quad (b)$$

$$\sum_j q_a^j(\bar{X}_jY_j)\bar{\eta}_j^+\bar{\eta}_j^+ = 0. \quad (c)$$